10. Define Conditional Probability. State and Prove multiplication theorem.

Solution :- Conditional Probability :- Probability of occurance of an event A (say) given that another event (B) (say) has already occurred is called Conditional probability, denoted by $P(A/B)$ and read as probability of A given B. It is defined as

$$P(A/B) = \frac{n(A \cap B)}{n(B)}, \quad \text{Similarly} \quad P(B/A) = \frac{n(A \cap B)}{n(A)}$$

obviously

$$P(A/B) = \frac{\frac{n(A \cap B)}{n(S)}}{\frac{n(B)}{n(S)}} = \frac{P(A \cap B)}{P(B)}. \text{ So}$$

$$P(A \cap B) = P(A/B)\, P(B).$$

This is called multiplication theorem.

* If the event A is independent from an event B then $P(A/B) = P(A)$.

* Multiplication ~~law~~ law of probability:-

$$P(A \cap B) = P(B)\, P(A/B).$$

Proof:- Let 'm' be favourable cases to the happening of an event B in 'n' trials, mutually exclusive & equally likely.

$$\therefore P(B) = \frac{m}{n} \quad \text{—— (1)}$$

But out of m outcomes favourable to this happening to B, let m_1 be favourable to the happening of the event A.

$\therefore$ Conditional probability of A, given that B has happened

$$P(A/B) = \frac{n(A \cap B)}{n(B)} = \frac{m_1}{m}$$

Now out of n trials, m_1 are favourable to the happening of A & B

$\therefore$ Probability of simultaneous occurence of A and B i.e., $P(A \cap B) = \frac{m_1}{n} = \frac{m_1}{m} \cdot \frac{m}{n}$

$$\boxed{P(A \cap B) = P(B) \cdot P(A/B)}$$

corollary:- If A and B are independent events, then $P(A/B) = P(A)$

$$\boxed{\therefore P(A \cap B) = P(A) \cdot P(B)}$$

11. In studying the causes of power failure, these data have been gathered: 5% are due to transformer damage, 80% are due to line damage, 1% involves both problems. Based on these percentages, approximate the probability that a given power failure involves

(i) Line damage given that there is a transformer damage $P(L|T)$

(ii) Transformer damage given that there is line damage $P(T|L)$

(iii) Transformer damage but not line damage $P(T\cap\bar{L})$

(iv) Transformer damage given that there is no line damage. $P(T|\bar{L})$

(V) Transformer damage or line damage.

Solution: — $P(A|B) = \frac{P(A\cap B)}{P(B)} \Rightarrow \underbrace{P(A\cap B) = P(B)\cdot P(A|B)}_{\text{Multiplication theorem}}$

$P(T)$: Power failure due to transformer damage

$P(L)$: " " " " Line "

Let T : Transformer damage causes power failure
L : Line damage causes power failure.

Given $P(T) = 0.05$, $P(L) = 0.80$, $P(T \cap L) = 0.01$

To find $P(L|T)$, $P(T|L)$, $P(T \cap \bar{L})$, $P(T|\bar{L})$ and $P(T \cup L)$.

(i) $P(L|T) = \frac{P(L \cap T)}{P(T)} = \frac{0.01}{.05} = .20$

(ii) $P(T|L) = \frac{P(L \cap T)}{P(L)} = \frac{0.01}{0.80} = 0.125$

(iii) $P(T \cap \bar{L}) = P(T) - P(T \cap L) = 0.05 - 0.01 = 0.04$

(iv) $P(T|\bar{L}) = \frac{P(T \cap \bar{L})}{P(\bar{L})} = \frac{0.04}{1-P(L)} = \frac{0.04}{1-0.8} = \frac{0.04}{.20} = 0.20$

(v) $P(T \cup L) = P(T) + P(L) - P(T \cap L)$
$= 0.05 + 0.80 - 0.01$
$= 0.84$

Notes

12. In a study of waters near power plants and other industrial plants that release waste water into the water system it was found that 5% showed signs of chemical and thermal pollution, 40% showed signs of chemical pollution, and 35% showed evidence of thermal pollution. Assume that the results of the study accurately reflect the general situation. What is the probability that a stream that shows some thermal pollution will also show signs of chemical pollution? What is the probability that a steam showing chemical pollution will not show signs of thermal pollution?

Solution :– Let C : water system showed signs of chemical pollution
T : water system showed signs of thermal pollution

Given: $P(C \cap T) = 0.05$, $P(C) = 0.40$, $P(T) = 0.35$

(i) $P(C|T) = \dfrac{P(C \cap T)}{P(T)} = \dfrac{0.05}{0.35} = 0.1429$

(ii) $P(\bar{T} \cap C) = P(C) - P(C \cap T) = 0.40 - 0.05$

$= 0.35$

(iii) $P(\bar{T}|C) = \dfrac{P(\bar{T} \cap C)}{P(C)} = \dfrac{0.35}{0.40} = 0.875$

Notas

(14) Define the Independance of two events. Consider the experiment of drawing a card from a well shuffled deck of 52 cards. The event A_1 denote drawing a spade card and the event A_2 denote an honor (10, J, K, Q, A). Are the events A_1 and A_2 are independent?

Solⁿ: — $P(A_1) = \frac{13}{52}$, & $P(A_2) = \frac{20}{52} = \frac{5}{13}$

$$P(A_1 | A_2) = \frac{5}{52} \Big/ \frac{5}{13} = \frac{13}{52} = P(A_1) \quad \text{Also } P(A_1 \cap A_2) = 5/52$$

$\Rightarrow P(A_1|A_2) = P(A_1)$ and $P(A_1 \cap A_2) = P(A_1) \cdot P(A_2) = \frac{13}{52} \times \frac{5}{13} = \frac{5}{52}$ $\Rightarrow$ A_1 & A_2 are independent.

(15) Let A and B be events such that $P(A) = 0.5$, $P(B) = 0.7$, what must be $P(A \cap B)$ equal for A and B to be independent?

Solⁿ: - For A and B to be independent, we must have $P(A \cup B) = P(A) \cdot P(B) = .5 \cdot .7 = .35$

(16) Let A and B be events such that $P(A) = 0.6$, $P(B) = 0.4$ and $P(A \cup B) = 0.8$. Are A and B independent?

Solution: - $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$$0.8 = 0.6 + 0.4 - P(A \cap B)$$

$$\Rightarrow P(A \cap B) = 1 - 0.8 = 0.2$$

$$P(A) \cdot P(B) = 0.4 \times 0.6 = .24$$

$$P(A \cap B) \neq P(A) \cdot P(B)$$

$\Rightarrow$ A and B are not independent.

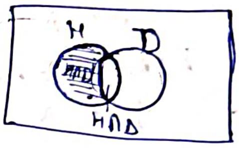
(17) The Probability that a unit of blood was donated by a paid donor is 0.67. If the donor was paid the probability of contracting serum hepatitis from the unit is 0.0144. If the donor was not paid, the probability of contracting serum hepatitis from the unit is 0.0012. A patient receives a unit of blood. What is the probability of the patient's contracting serum hepatitis from the source?

Solution: - Let D: a unit blood was donated by a paid donor,

H: a person contracts hepatitis from a unit of blood.

Given $P(D) = 0.67$, $P(H|D) = \frac{P(H \cap D)}{P(D)} = 0.0144$

Thus, the probability that a unit of blood was donated by a not paid donor is



$P(D') = 1 - P(D) = 1 - 0.67 = 0.33.$

$P(H|D') = 0.0012$

$\frac{P(H \cap D')}{P(D')} = 0.0012$

$P(H \cap D') = 0.0012 \times 0.33$

$= 0.000396$

$P(H \cup D) = \cancel{0.000396} \; \cancel{+ 0.67}$

$= 0.670396$

$\cancel{P(H \cup D') = P(H) + P(D') - P(H \cap D')}$

$\cancel{0.000396 = P(H) + 0.33 -}$

$\cancel{P(H \cap D) =}$

$\cancel{P(H \cup D) = P(H) + P(D) - P(H \cap D)}$

$\cancel{P(H \cup D') = P(H) + P(D') - P(H \cap D')}$

$P($

$P(H \cap D) = .009648$

$P(H) = P(H \cap D) + P(H \cap D') = P(H|D)\,P(D) + P(H|D')\,P(D')$

$= 0.0144 \times 0.67 + 0.0012 \times .33$

$= 0.010044.$

(20) The use of plant appearance in prospecting for ore deposits is called geobotanical prospecting. One indicator of copper is a small mint with a mauve - colored flower. Suppose that, for a given region, there is a 30% chance that soil has a high copper content and 23% chance that the mint will be present there. If the copper content is high, there is a 70% chance that the mint will be present.

(i) Find the probability that the copper content will be high and the mint will be present.

(ii) Find the probability that the copper content will be high given that the mint is present.

Sol:- Let C: Soil has high copper content and M: Soil has some mint content

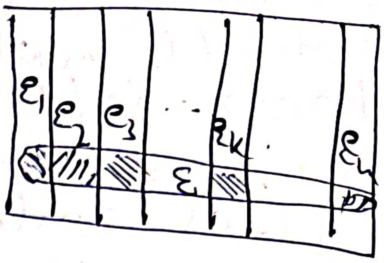
Given $P(C) = 0.3$ $\quad P(M) = .23$, $P(M|C) = .7$

(i) $P(M \cap C) = P(M|C)\,P(C) = 0.7 \times .3 = 0.21$

(ii) $P(C|M) = P(C \cap M) / P(M) = 0.21 / .23$

$= 0.9130$

Total Probability Theorem (T.P.T) :- Let $E_1, E_2, \dots E_n$ are n. M.E and set of exhaustive events and E is also defined in the same sample space.



$P(E) = P(E\cap E_1) + P(E\cap E_2) + \dots + P(E\cap E_n)$

$P(E) = \sum_{r=1}^{n} P(E\cap E_r)$

$P(E|E_r) = \dfrac{P(E\cap E_r)}{P(E_r)}$

$\boxed{P(E) = \sum_{r=1}^{n} P(E_r)\,P(E|E_r)}$ $\quad P(E\cap E_r) = P(E|E_r)\,P(E_r)$

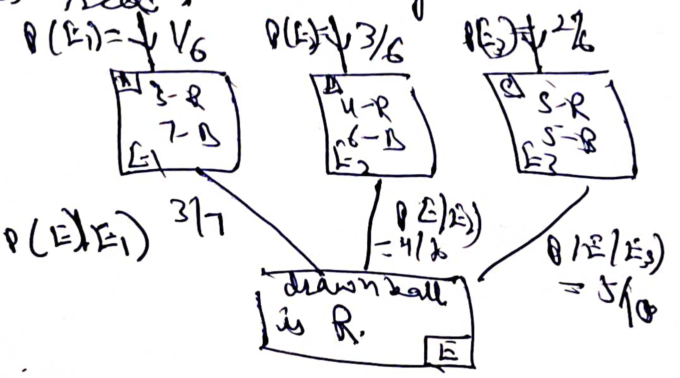
$P(E) = P(E_1)P(E|E_1) + P(E_2)P(E|E_2) + \dots + P(E_n)P\left(\frac{E}{E_n}\right)$

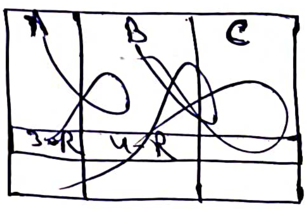
Probability that 1 is used

P. " of E through E_1

Q:- Let 3 boxes A, B & C are given where A contains 3-R & 7-B, B contains 4-R & 6 B and C contains 5-R & 5-B. Now a die is rolled and if outcome is less than 2 then 1 ball from A is selected or if outcome is 2, 3, 4, then 1-ball from B is selected otherwise 1-ball from C is selected. Then find the probability that selected ball is red.

Soln:-





$$P(E) = P(E_1)P(E|E_1) + P(E_2)P(E|E_2) + P(E_3)P(E|E_3)$$

$$= \frac{1}{\cancel{6}\,2}\times\frac{\cancel{3}}{10} + \frac{\cancel{3}}{\cancel{6}\,2}\times\frac{\cancel{4}\,2}{\cancel{10}\,5} + \frac{\cancel{2}}{6}\times\frac{\cancel{5}}{\cancel{10}\,2}$$

$$= \frac{1}{20} + \frac{1}{5} + \frac{1}{6}$$

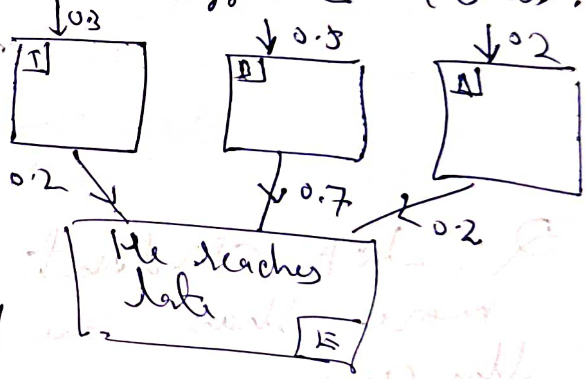
$$\frac{3 + 12 + 10}{60} = \frac{\cancel{25}\,5}{\cancel{60}\,12}$$

$$= 5/12$$

2	20, 5, 6
2	10, 5, 3
5	5, 5, 3
3	1, 1, 3
	1, 1, 1

Q:- A goes from home to office using train, bus or auto with respective probability 0.3, 0.5 & 0.2. If he uses train the probability that he will be late is 0.2 and the corresponding probability for bus and auto are 0.7 and 0.2 respectively. Now find the probability that he reaches office on time.

Sol^n:- -



$$P(E) = 0.3\times.2 + 0.5\times0.7 + 0.2\times0.2$$

$$= 0.6 + .35 + .04$$

$$= 0.45$$

Probability that he reaches on time

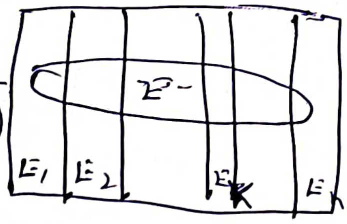
$$= 1 - P(E)$$

$$= 1 - 0.45$$

$$= 0.55$$

Baye's theorem :—

$$P(E_k|E) = \frac{P(E_k \cap E)}{P(E) = \sum_{i=1}^{n} P(E_i) P(E|E_i)}$$



$$P(E_k|E) = \frac{P(E_k) \cdot P(E|E_k)}{P(E) = \sum_{i=1}^{n} P(E_i) P(E|E_i)}$$

P (that batch E_k was used given E has occured)

$$= \frac{\text{contribution of batch}}{\text{Total}}$$

(It was found, it was seen)

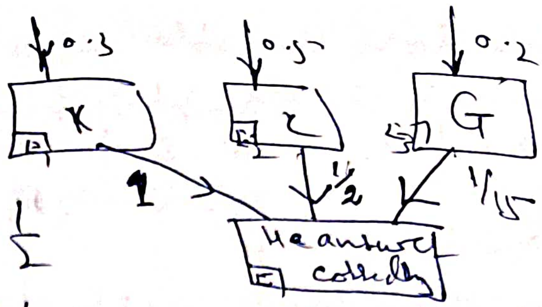
Q:– If it was found to be red. Now, find the probability that bag A was used

$$P(E_1|E) = \frac{P(E_1) P(E|E_1)}{P(E) = \sum_{i=1}^{n} P(E_i) P(E|E_i)}$$

$$= \frac{\cancel{\frac{2}{7}} \times \frac{1}{2}}{5/12} = \frac{1}{\cancel{14}_{7}} \times \frac{\cancel{12}^{6}}{5} = \frac{6}{35}$$

Q:– A student attempts a M.C.Q (with more than one correct). He either knows the answer or copies it or make a guess. with respective probabilities as 0.3, 0.5 & 0.2. Also the probability that his answer is correct if he copies is 1/2. Now it was found that his answer is correct. find the probability that he knew the answer.

Sol^n:-



$$P(E) = 1 \times 0.3 + 0.5 \times \frac{1}{2} + 0.2 \times \frac{1}{15}$$

$$= 0.3 + 0.25 + \frac{0.2}{15} = \frac{6.2}{15}$$

$${}^4C_1 + {}^4C_2 + {}^4C_3 + {}^4C_4 = 2^4 - 1 = 15$$

$$P(E_K|E) = \frac{P(E_K)\, P(E|E_K)}{P(E)} = \frac{0.3 \times 1}{\frac{6.2}{15}} = \frac{45}{62}$$

Baye's Theorem :-(statement) If $B_1, B_2, \ldots B_n$ are mutually exclusive & exhaustive events with ~~P(B)~~ $P(E_i) \neq 0$ $(i = 1, 2, \ldots n)$ of a random experiment then for any arbitary event E of the sample space with $P(E) > 0$, we have

$$P(E_i/E) = \frac{P(E_i)\, P(E/E_i)}{\sum_{i=1}^{n} P(E_i)\, P(E/E_i)}$$